DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
UNIVERSITY OF BARISHAL
FINAL EXAMINATION-2022
Course Title: Artificial Intelligence
Course Code: CSE-3205
3rd Year 2nd Semester
Admission Session: 2018-19

Time: 3 hours **Marks:** 60

Answer any five Questions from the followings.

1. a) What is AI? Differentiate between the strong AI and Weak AI. [3]

b) Identify the four sources that demonstrate Evidence of Human Intelligence. [3]

c) Distinguish between the following properties of a task environment: [3]
   i. Static or dynamic
   ii. Discrete or continuous
   iii. Single or multi-agent

d) Briefly explain the PEAS description of the task environment for **an automated taxi.** [3]

2. a) Describe the function of vacuum cleaner as an Agent with suitable diagram. [3]

b) Convert the following sentence into clausal form: [4]

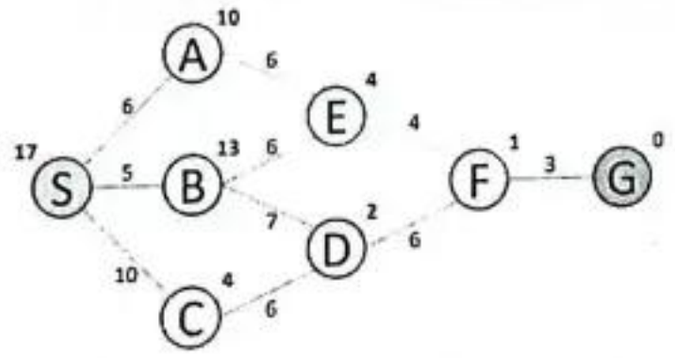
$$\exists x \forall y (\forall z P(f(x), y, z) \rightarrow (\exists u\, Q(x, u)\ \&\ \exists v\, R(y, v)))$$

c) For the following database facts, write a completion formula that states that Johny is the only person that lives in Barishal. [5]

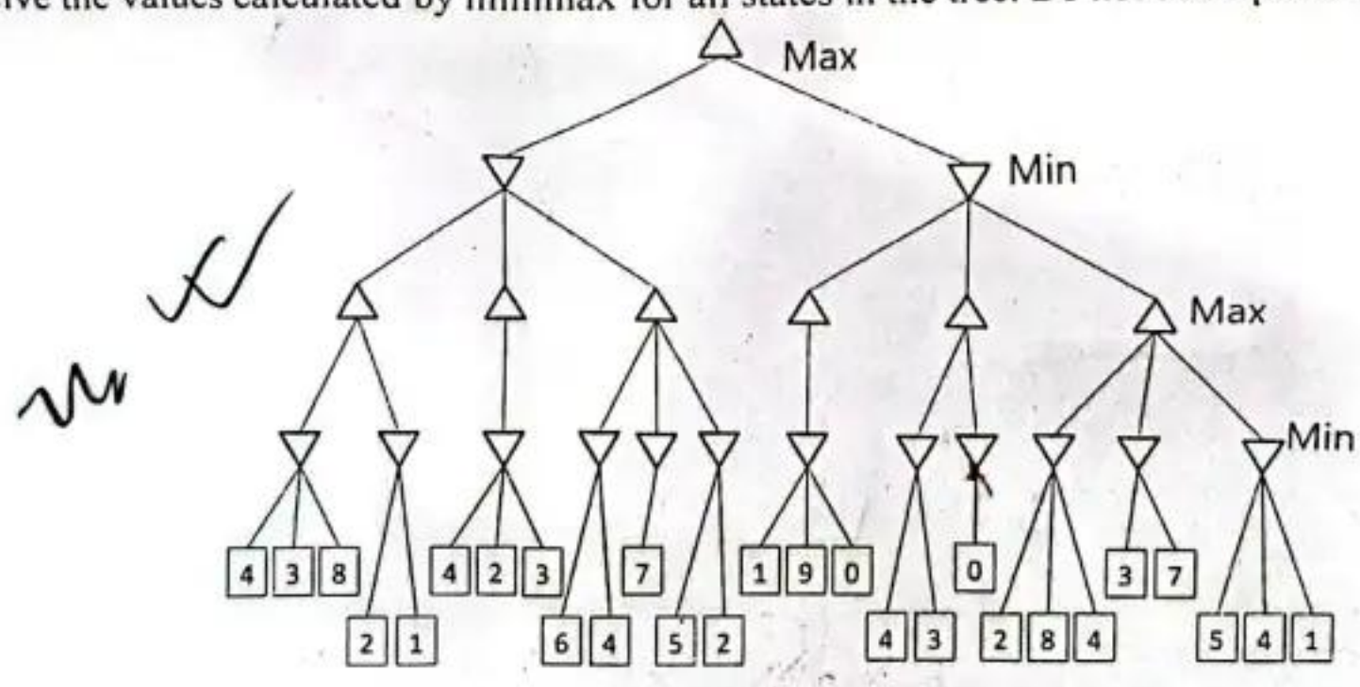
LIVESIN (johny, Barishal)
LIVESIN(alice, Pabna)
LIVESIN(darth, bogura)
OWNS(johny, smartphone)
STUDENT(darth)

3. a) Define the following terms in your own words: state, state space, search tree, search node, goal, action, transition model, and branching factor. [4]

b) Perform A* Algorithm on the following figure. Explicitly write down the queue at each step [8]



4. a) Give the values calculated by minimax for all states in the tree. Do not use alpha-beta pruning. [6]



b) Indicate which branches of the tree will be pruned by alpha-beta pruning? [6]

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